

1. WHAT IS ARTIFICIAL INTELLIGENCE?

One of the fascinating aspects of the field of artificial intelligence (AI) is that the precise nature of its subject matter turns out to be surprisingly difficult to define. The problem, of course, has two parts, since securing an adequate grasp of the nature of the artificial would do only as long as we were already in possession of a suitable understanding of the idea of intelligence. What is supposed to be “artificial” about artificial intelligence, no doubt, has to do with its origins and mode of creation in arising as a product of human contrivance and ingenuity rather than as a result of natural (especially biological or evolutionary) influence. Things that are *artificially intelligent*, in other words, differ from those that are *naturally intelligent* as artifacts that possess special properties ordinarily possessed by non-artifacts. So these are things that have a certain property (intelligence) as a result of a certain process (because they were created, designed, or manufactured in this way).

These artifacts, of course, are of a certain special kind in the sense that they are commonly thought of as being *machines*. If machines are simply things that are capable of performing work, then, since human beings are capable of performing work, human beings turn out to be machines, too. If anyone wants to know whether or not there are such things as intelligent machines, therefore, the answer seems obvious so long as human beings are (at least, some of the time) *intelligent*. A more difficult question, whose answer is less evident, however, arises from distinguishing between animate and inanimate machines. Human beings, after all, may result from human contrivance and ingenuity, but they are clearly biological in their origin. The issue thus becomes whether or not *inanimate* machines, as opposed to human beings, are capable of possessing a certain special property that human beings are supposed to display – if not always, at least on certain special occasions.

The problem revolves about the identification or the definition of what is meant to be “intelligent” about artificial intelligence. The dictionary may seem to be an appropriate place to begin. *Webster’s New World Dictionary* (1988), for example, defines “intelligence” as “(a) the ability to learn or understand from experience; ability to acquire and retain knowledge; mental ability; b) the ability to respond quickly and successfully to a new situation; use of the faculty of reason in solving problems, directing

conduct, etc., effectively; c) in *psychology*, measured success in using these abilities to perform certain tasks". If the concept of intelligence is to be applicable (at least, in principle) to inanimate machines, however, it must not be the case that inanimate machines could not possibly be intelligent as "a matter of definition".

Matters of definition, of course, are often thought to be arbitrary or capricious, capable of resolution merely by mutual agreement: "The question is who is to be master, that's all!" But much more hinges upon the meaning that we assign to the words that we use when we address specific problems and search for suitable answers. In this case, the establishment of an appropriate definition for "intelligence" becomes equivalent to a theory of the nature of intelligence as a phenomenon encountered in the world. Hence, for example, any analysis that has the effect of *excluding* human beings from the class of intelligent things, on the one hand, or that has the effect of *including* tables and chairs within the class of intelligent things, on the other, would thereby display its own inadequacy. By reflecting upon "exemplars" of the property in question – instances in which its presence or absence and other features these things possess are not in doubt – it should be possible to achieve conceptual clarification, if not to secure complete agreement, on the crucial features of intelligence itself, which amounts to a prototype theory.

Even if we assume that human beings *are* among the things that happen to be intelligent, the problem remains of isolating those specific aspects of human existence that are supposed to be "intelligent". Since human beings exhibit anger, jealousy, and rage, it might be asked if inanimate machines would be intelligent if they could exhibit anger, jealousy, and rage. Notice, the point is not the trivial one that we are likely to mistake anger, jealousy, and rage for "intelligence" as a matter of fact, but rather the subtle one that *unless we already know that anger, jealousy, and rage are not aspects of "intelligence"*, we are not able to rule them out. If human beings are the best exemplars of the property in question, yet are widely observed to exhibit anger, jealousy, and rage, then by what standard, yardstick, or criterion can we tell that these are *not* some of the features characteristic of intelligence? And, in fact, there are situations, circumstances, and conditions under which anger, jealousy and rage would be intelligent. So how can we possibly tell?

While these definitions may seem helpful with respect to human beings, they remain problematic in application to machines. If we take for granted that human beings do possess the ability to learn or understand from